Adversarial Attacks

Robust Deep Learning (Seminar)

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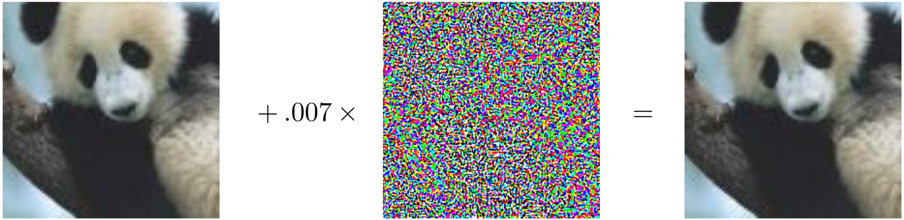
Two images

Two images

panda

gibbon

A fragile foundation



$\boldsymbol{x}$
"panda"
57.7% confidence

$\text{sign}(\nabla_{\boldsymbol{x}} J(\boldsymbol{\theta}, \boldsymbol{x}, y))$
"nematode"
8.2% confidence

$\boldsymbol{x} + \epsilon\text{sign}(\nabla_{\boldsymbol{x}} J(\boldsymbol{\theta}, \boldsymbol{x}, y))$
"gibbon"
99.3 % confidence

Goodfellow et al., 2014

Transferability

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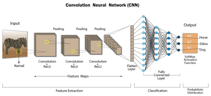

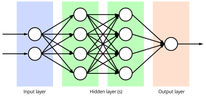

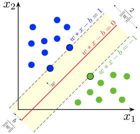

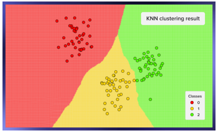

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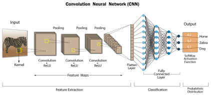

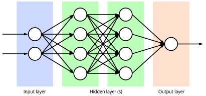

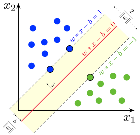

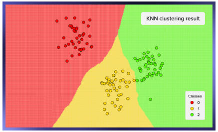

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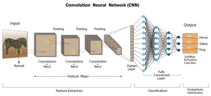

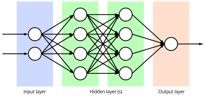

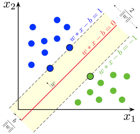

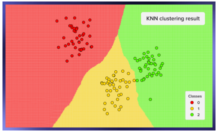

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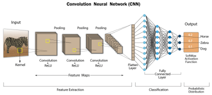
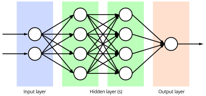
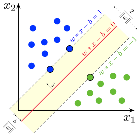
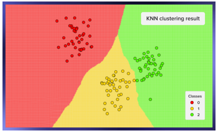
$\leadsto$ The attacker needs **no access** to the target. Train your own model, attack that, and the perturbation carries over.









Why this matters

- What we just saw exposes a **fundamental blind spot** — the model and the human are not looking at the same thing.

Images: medicalimaging.na; evsmarts.com; thetravelimages.com

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- That raises hard questions about the **reliability** of systems we already treat as dependable.

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Why this matters

- What we just saw exposes a **fundamental blind spot** — the model and the human are not looking at the same thing.
- That raises hard questions about the **reliability** of systems we already treat as dependable.
- And it grows more pressing as these systems are woven ever deeper into **everyday life**, where failures carry real consequences.

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Outline

1. What is an adversarial example?

2. Taxonomy

3. Evasion attacks

4. Poisoning attacks

5. Discussion

5.1 Physical adversarial attacks
5.2 Real-life adversarial attacks
5.3 Explaining adversarial examples
5.4 Adversarial attacks as a subject of study

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Definition

Definition 1 (Adversarial Example)

An *adversarial example* is an **input to a machine learning model** that has been **intentionally perturbed**, typically **in a subtle and often imperceptible manner**, such that the model produces an incorrect prediction while the semantic content of the input remains unchanged.

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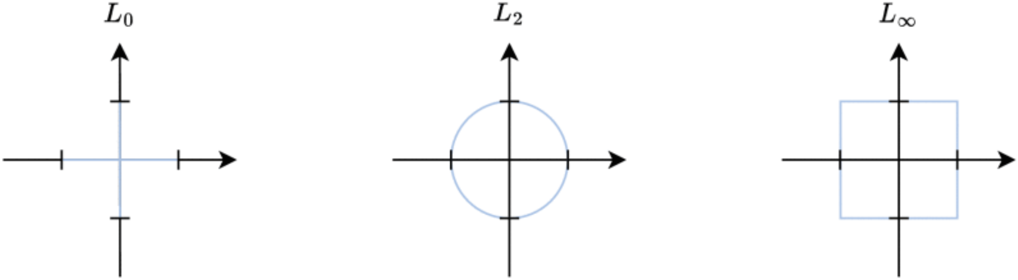
- **Machine learning models** — no quirk of deep networks, but a general property of machine learners.
- **Intentional perturbations** — generated systematically through optimisation and exploited by an adversary, not stumbled upon.
- **Subtle and imperceptible perturbations** — mathematically small under a chosen norm, and visually indistinguishable to humans; the latter is asserted far more often than it is rigorously validated.

Measuring “small”: the ℓ_p norms



$||x||_0 = (|x_1|^0 + \ldots + |x_n|^0)$ $||x||_2 = (|x_1|^2 + \ldots + |x_n|^2)^{\frac{1}{2}}$ $||x||_\infty = \max_i |x_i|$

The norm defines the **perturbation budget** ϵ and, with it, what “similar” even means.

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Four orthogonal axes

- **Attacker knowledge** — white-box vs. black-box
- **Attack stage** — evasion (test time) vs. poisoning (train time)
- **Attack goal** — targeted vs. untargeted
- **Perturbation constraint** — which norm, which budget

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L-BFGS

White-box · Evasion · Targeted · ℓ_2

The arc of the field

2013 **L-BFGS** — expensive, but proves existence

DeepFool

White-box · Evasion · Untargeted · ℓ_2

The arc of the field

2013 **L-BFGS** — expensive, but proves existence

2016 **DeepFool** — minimal perturbation via linear approximation

Carlini & Wagner

White-box · Evasion · Targeted · $\ell_0/\ell_2/\ell_\infty$

The arc of the field

2013 **L-BFGS** — expensive, but proves existence

2016 **DeepFool** — minimal perturbation via linear approximation

2017 **C&W** — reformulate as optimisation; breaks defences

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Physical adversarial attacks

Real-life adversarial attacks

Explaining adversarial examples

Adversarial attacks as a subject of study

Take-home messages

References I

Backup: attack comparison